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Epoch 29/40
1875/1875 [=====] - 3s 2ms/step - loss: 5.1357 - acc: 0.8942 - val_loss: 6.1563 - val_acc: 0.8961
Epoch 30/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.1226 - acc: 0.8942 - val_loss: 6.0663 - val_acc: 0.8972
Epoch 31/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.0476 - acc: 0.8953 - val_loss: 6.3820 - val_acc: 0.8876
Epoch 32/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.1187 - acc: 0.8938 - val_loss: 6.5461 - val_acc: 0.8898
Epoch 33/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.0417 - acc: 0.8949 - val_loss: 6.2389 - val_acc: 0.8965
Epoch 34/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.1899 - acc: 0.8936 - val_loss: 6.0297 - val_acc: 0.9060
Epoch 35/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.1787 - acc: 0.8945 - val_loss: 6.2162 - val_acc: 0.8983
Epoch 36/40
1875/1875 [=====] - 3s 2ms/step - loss: 5.1070 - acc: 0.8950 - val_loss: 6.2734 - val_acc: 0.8943
Epoch 37/40
1875/1875 [=====] - 3s 1ms/step - loss: 4.9866 - acc: 0.8953 - val_loss: 6.4508 - val_acc: 0.8888
Epoch 38/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.1557 - acc: 0.8949 - val_loss: 6.9689 - val_acc: 0.8829
Epoch 39/40
1875/1875 [=====] - 3s 1ms/step - loss: 5.1465 - acc: 0.8952 - val_loss: 6.3960 - val_acc: 0.9002
Epoch 40/40
1875/1875 [=====] - 3s 1ms/step - loss: 4.9815 - acc: 0.8967 - val_loss: 6.2203 - val_acc: 0.8968
110.3399152757373

```

Figure 3: Python takes 110.3 seconds in a neural network

It contains an input layer, hidden layer, and output layer. To avoid complexities, we did not use any preprocessing on our data set, and worked on it as it is. We trained this model for 40 iterations and checked the time difference between Julia and Python in it.

For Julia, the Flux library was used to implement the neural network and for Python, the Keras library was used. Figure 2 shows the time taken by Julia in a neural network.

Figure 3 shows the time taken by Python and a few iterations of the model in the neural network.

The results show that there are huge time differences between Julia and Python when it comes to a neural network.

Table 1 summarises all the results of our experiments, and gives the time difference (in percentage) between Julia and Python.

All the experiments we carried out indicated that as the complexity of the program as well as the size of the data set increases, the execution time difference between Julia and Python increases. From the results, we can conclude that Julia is the better programming language for machine learning and neural networks. 

Experiment	Julia (seconds)	Python (seconds)	Time difference (%)
Matrix multiplication (without library)	0.000001	0.0015	99.9
Matrix multiplication (with library)	0.000017	0.0013	98.69
Linear search (using loop)	0.42	16.4	97.43
Linear search (using IN operator)	0.43	6.2	93.06
Linear regression	0.000718	0.00718	90
Logistic regression	0.025	0.34901	92.83
Neural networks	5.76	110.3	94.77

Table 1: Time difference between Julia and Python for all experiments

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